Central Limit Theorem

1 RECOGNISE IT

- Population shape is not normal or is unknown.
- Probability concerns a sample mean.
- Asked whether $\bar{X}$ is approximately normal for a stated n.

2 DECISION / METHOD

1. Identify the population shape.
2. If population normal: $\bar{X}$ normal for any n.
3. Official guide: fairly symmetric often n>15; most distributions n>30.
4. Use centre μ and SE σ/√n.

3 FORMULA

for sufficiently large n:

$\bar{X}$ approx $N(\mu, \sigma^2/n)$

4 FORMULA SHEET VS WHAT I KNOW

Gives: SE formula under conditions.
Still know: CLT concerns the sampling distribution of $\bar{X}$, not the raw population becoming normal.

5 EXCEL ROUTE

Once justified, use:
`=NORM.DIST(boundary,mu,sigma/SQRT(n),TRUE)`
with complements/differences as needed.

6 TINY WORKED EXAMPLE

A skewed population has μ=20, σ=10 and n=100. CLT gives $\bar{X}$ approximately $N(20,1^2)$.

7 TOP EXAM TRAPS

- Claiming raw observations become normal.
- Ignoring strong skew with small n.
- Forgetting that SE shrinks as n grows.

8 SAY ALOUD

The CLT makes sample means approximately normal for large n, while keeping centre μ and spread σ/√n.

9 EXIT CHECK

Does the CLT make the original population normal?

ONE-GLANCE VISUAL ROUTE

